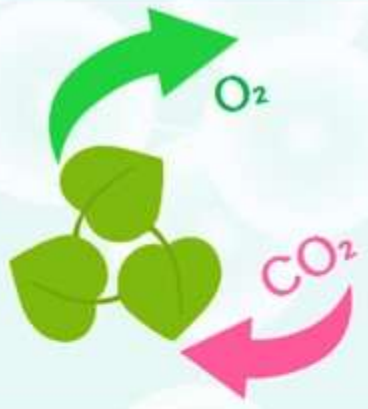


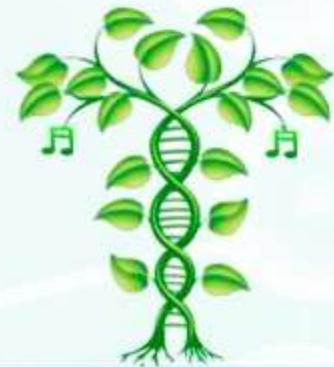


# BIOLOGY



## TISSUES

MODULE - 8



## WHAT DOES A LIVING THING COMPRISES OF?



ORGANISM



ORGAN SYSTEM



ORGANS



TISSUES



CELLS

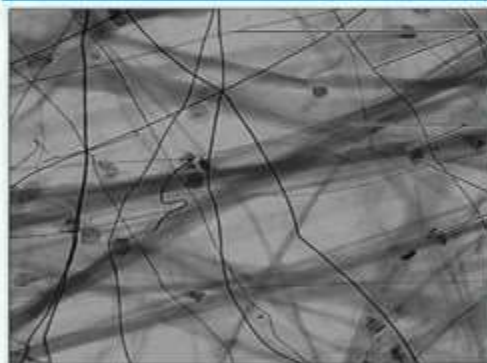


# ANIMAL TISSUE

EPITHELIAL



CONNECTIVE



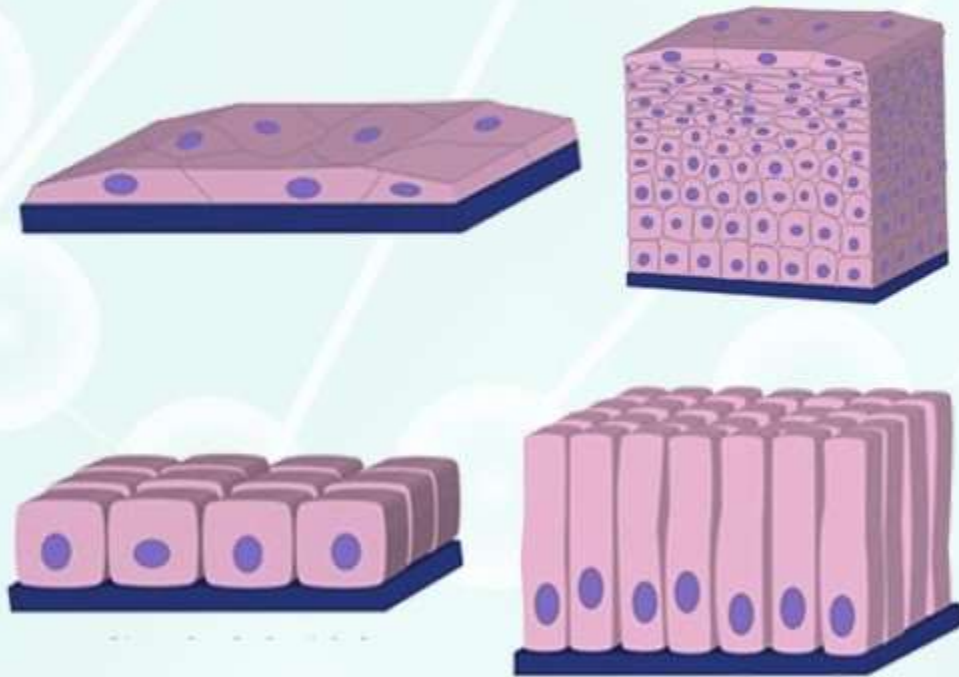
MUSCULAR



NERVOUS



## EPITHELIAL TISSUE

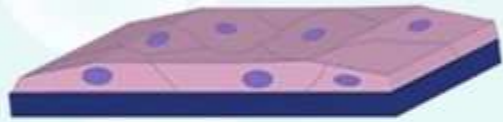


- Epithelial tissues line all the organs in the body.
- They have high regenerative capacity and high reproduce rapidly.
- Their main functions involve
  - Protection
  - Secretion
  - Absorption
  - Sensation



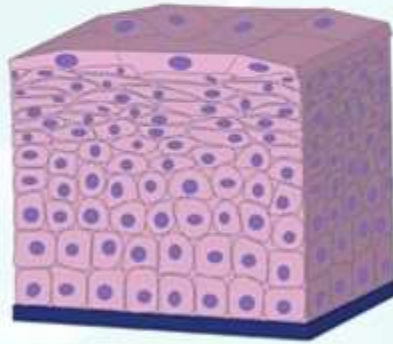
# EPITHELIAL TISSUE

## SQUAMOUS



- Heart
- Blood vessels
- Lymphatic vessels

## STRATIFIED



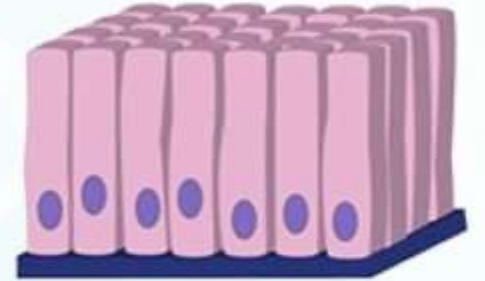
- Esophagus
- mouth

## CUBOIDAL



Kidney tubules

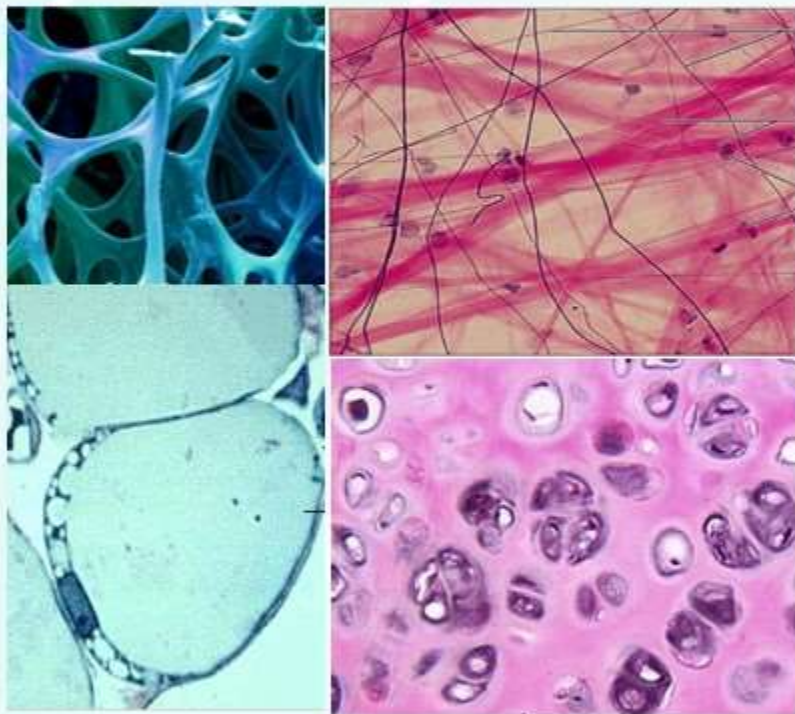
## COLUMNAR



- Digestive tract
- uterus



## CONNECTIVE TISSUE

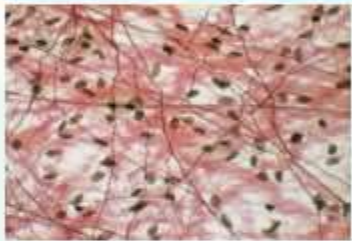


- Connective tissue maintains the form of the body and the organs and improves cohesion and internal support.
- All forms of connective tissue are composed of
  - extracellular fibers,
  - an amorphous matrix called ground substance,
  - stationary and migrating cells.



# CONNECTIVE TISSUE

## AREOLAR



- White collagen fibres
- Yellow elastic fibres

## FIBROUS



Tendon



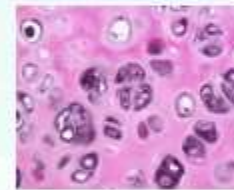
Ligament

## ADIPOSE



Adipocytes

## SKELETAL



- Bone
- Cartilage

## FLUID



Blood

Lymph

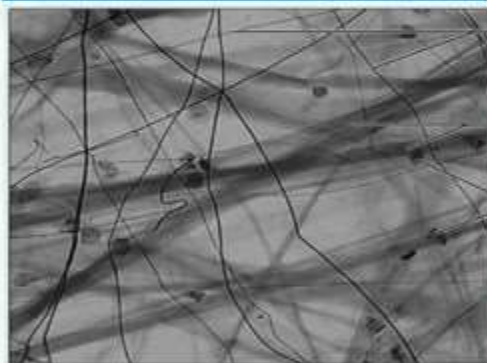


# ANIMAL TISSUE

EPITHELIAL



CONNECTIVE



MUSCULAR

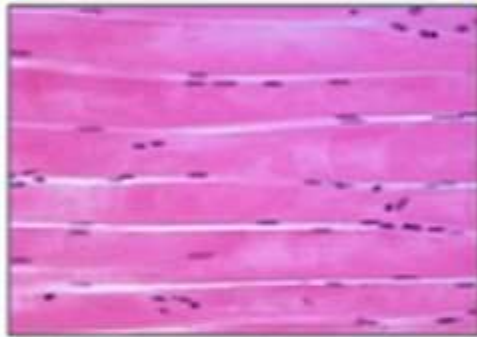


NERVOUS



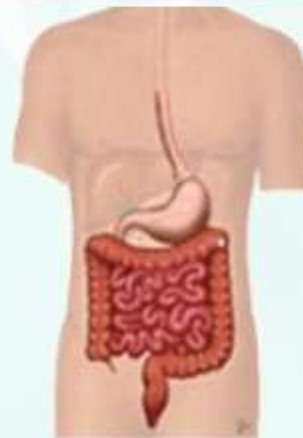
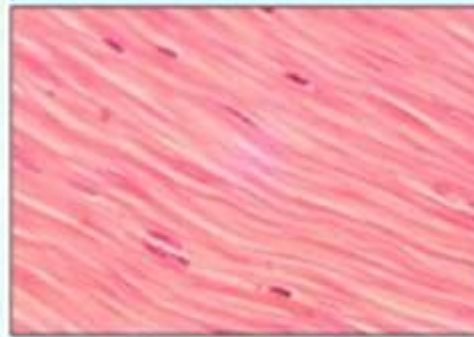
# MUSCLE TISSUES

## SKELETAL



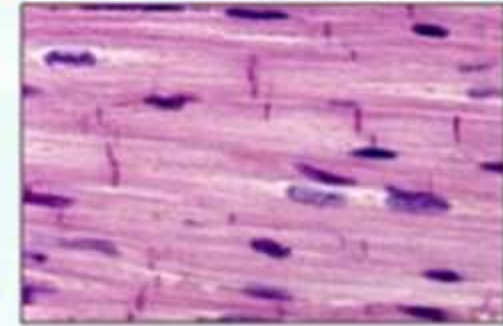
- Limbs
- Tongue
- Face
- Neck

## SMOOTH



- Ureters
- Bronchi
- Alimentary canal
- Blood vessels

## CARDIAC



Walls of Heart



## MUSCLE TISSUES

### SKELETAL

1. Skeletal muscles / voluntary muscles/ striped/ striated muscle fibres
2. Long cylinder shaped unbranched cells.
3. Multinucleated
4. Have dark and light bands
5. Contract quickly but cant remain contracted for long. So, gets tired /fatigued

### CARDIAC

1. Cardiac/ smooth muscle fibres
2. Short cylinder shaped branched cells
3. Uninucleated
4. Faint dark and light bands
5. Contract quickly and rhythmically. So never gets fatigued/ tired

### SMOOTH

1. Smooth Muscles/ involuntary muscles/ unstriped / non- striated muscle fibres.
2. Spindle shaped cells
3. Uninucleated
4. No dark and light bands
5. Contract slowly but can remain contracted for long. So don't get fatigued/ tired

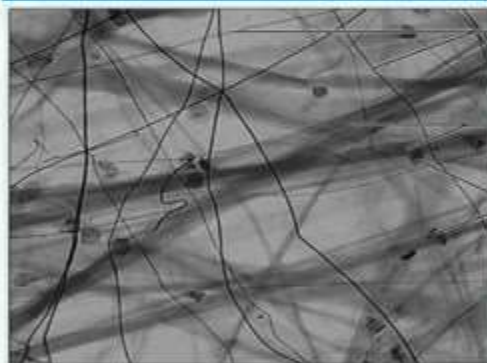


# ANIMAL TISSUE

EPITHELIAL



CONNECTIVE



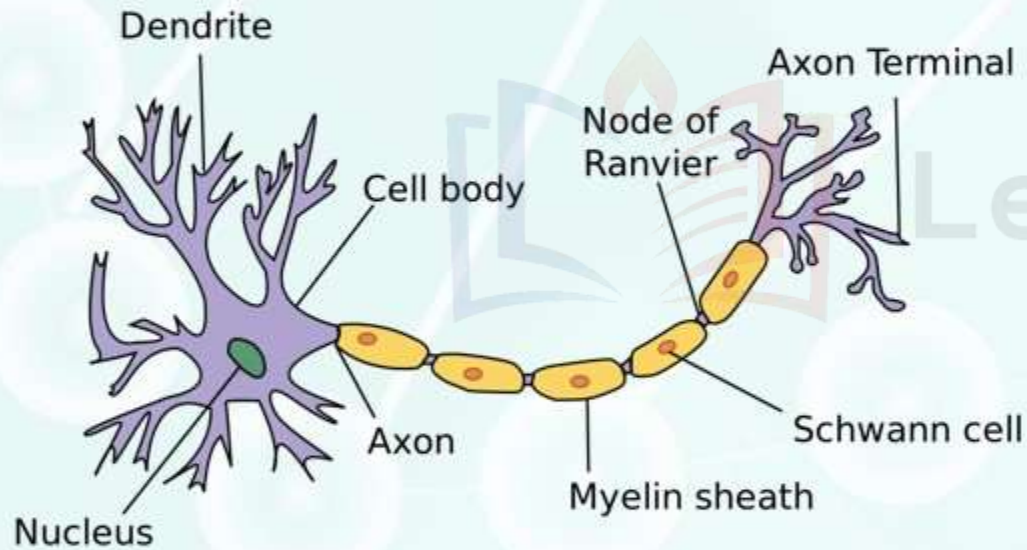
MUSCULAR



NERVOUS

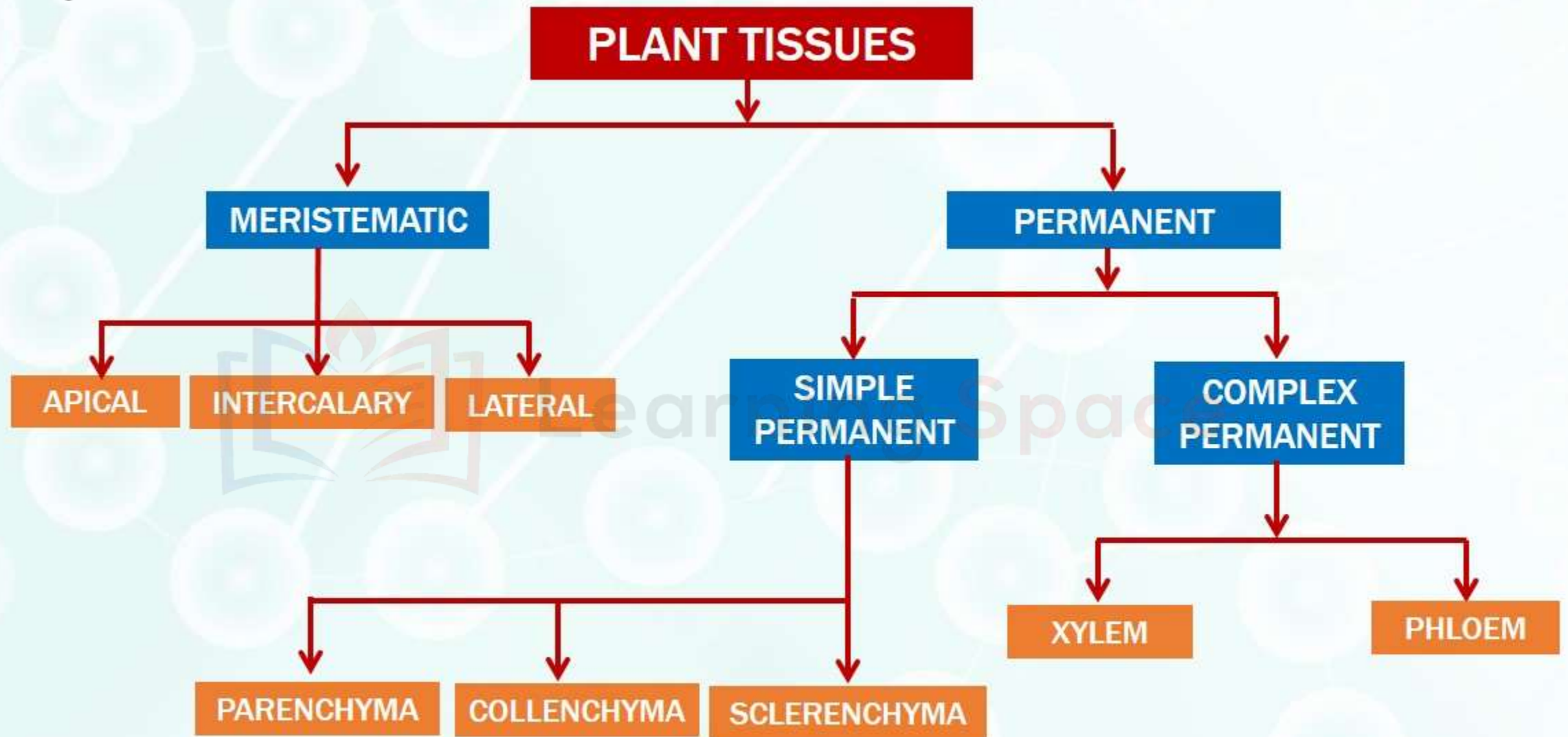


## NERVOUS TISSUES

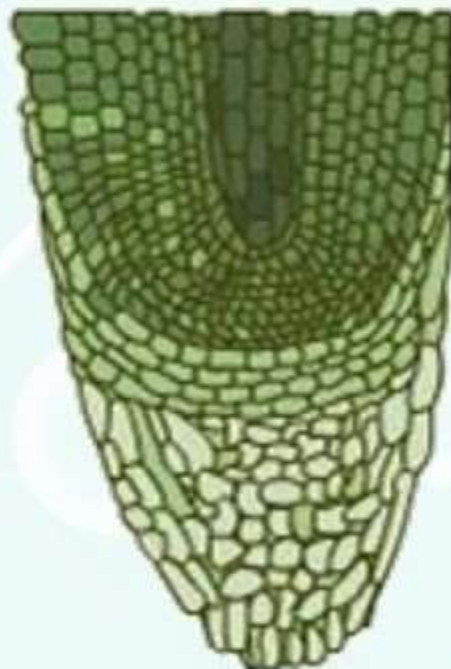


- They are highly specialized for being stimulated and then transmitting the stimulus very rapidly from one place to another.
- Brain, spinal cord and nerves are made up of nervous tissue.
- Nervous tissue contain unit cell called nerve cell or neurons and some glial cells.
- The impulses travel from one neuron to another neuron.
- Neural cells can't divide because they lack centrioles.





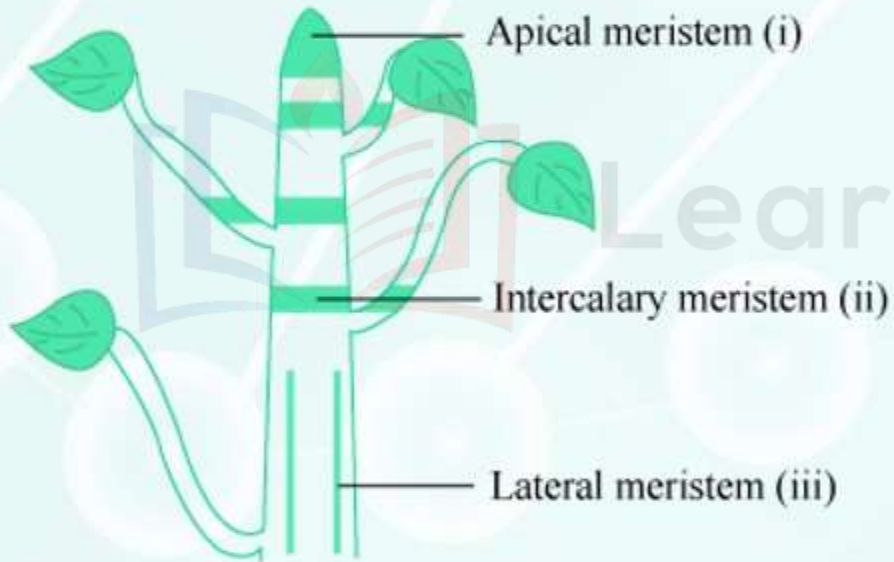
## MERISTEMATIC TISSUES



- A meristematic tissue constitutes a group of actively dividing cells present in the growing region of plant, e.g., the tips of roots and stems.
- The cells of tissue are compactly arranged and do not have intercellular space.
- The cells have dense protoplasm with prominent nuclei.
- Vacuoles in these cells are either small or absent.



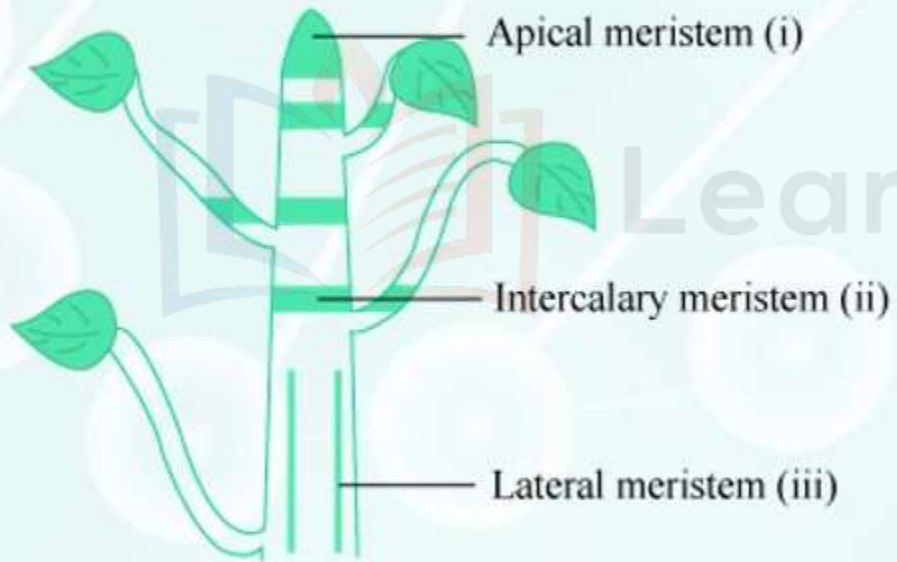
## MERISTEMATIC TISSUES



- Apical meristems are present at the tips of roots, shoots, branches and leaves.
- It brings about the elongation of the root and stem. It results in increase in the height of the plant, which is called primary growth.



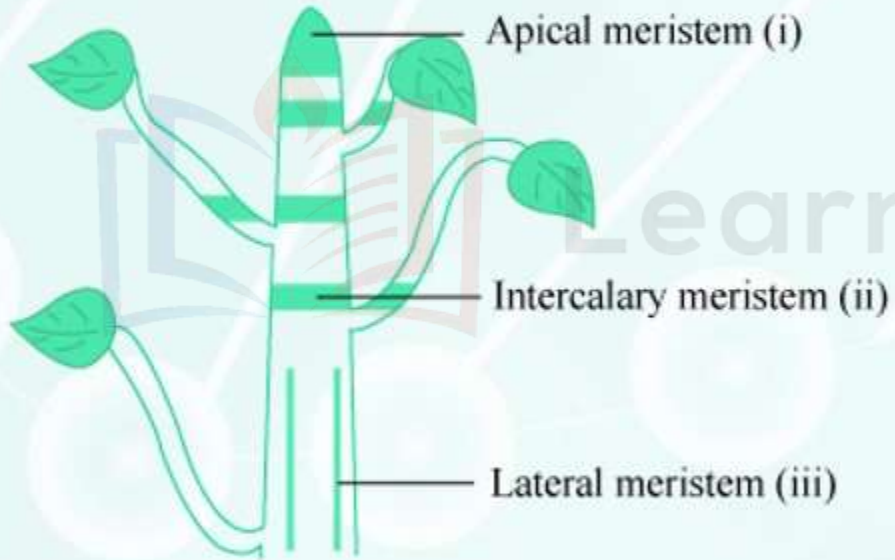
## MERISTEMATIC TISSUES



- Intercalary meristem are located at the base of leaves or internodes.
- These are found in stems of grasses and other monocots or below the nodes (as in mint etc.)
- It produces an increase of length of organ such as leaves and internodes.



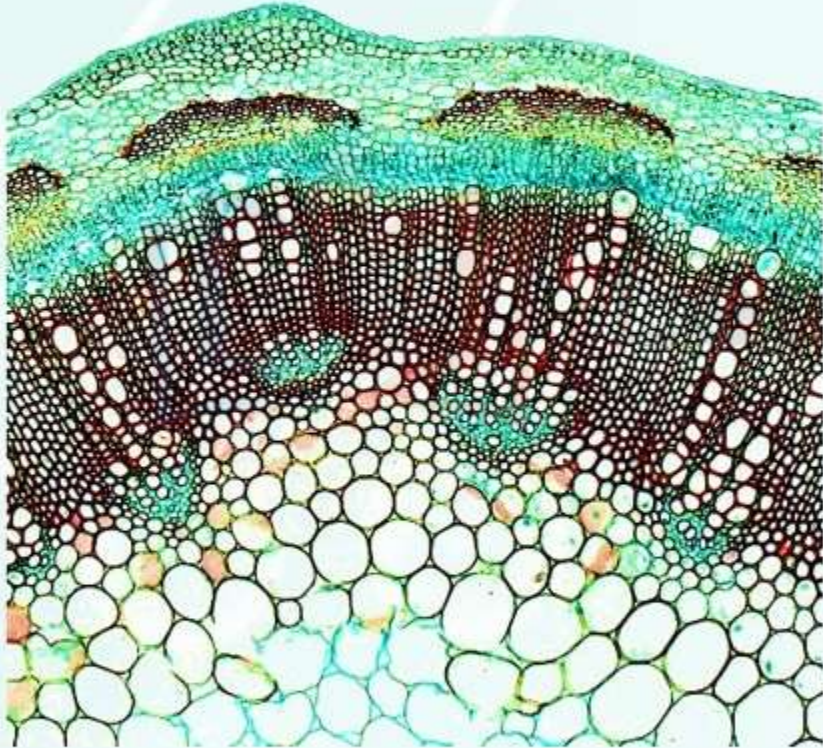
## MERISTEMATIC TISSUES



- Lateral meristems are present along the lateral side of the stems and roots. For growth of bark.
- It causes the organ (stem or root) to increase in diameter and girth. This is called secondary growth.



## PERMANENT TISSUES



- A permanent tissue is a group of cells, which is derived from the meristematic tissues, **but these cells have lost the power of division**
- The development process by which cells that have been derived from meristematic tissue, take up a permanent shape, size and function is called **differentiation**.
- Permanent tissues are of two types
  - **Simple and**
  - **Complex permanent tissue.**



# SIMPLE PERMANENT TISSUES

## PARENCHYMA

- Parenchyma forms the bulk of plant body. It consists of thin walled living cells.
- The cell wall is thin and encloses a dense cytoplasm which contains a small nucleus and a large central vacuole.
- The main function of parenchymatous tissue is storage of food, e.g., starch in the parenchyma of cortex of potato tuber.
- Parenchyma serves as packing tissue to fill the spaces between other tissues.

## COLLENCHYMA

- Collenchyma is usually found in 3-4 layers beneath epidermis in stem, petioles and leaves of herbaceous plants.
- In collenchyma, intercellular spaces are generally absent.
- It provides the mechanical support, protection, flexibility and elasticity to the plants organs.
- It allows easy bending in various parts of the plant (leaf, stem) without breaking.
- When cells of collenchyma contain some chloroplasts, they manufacture sugar and starch

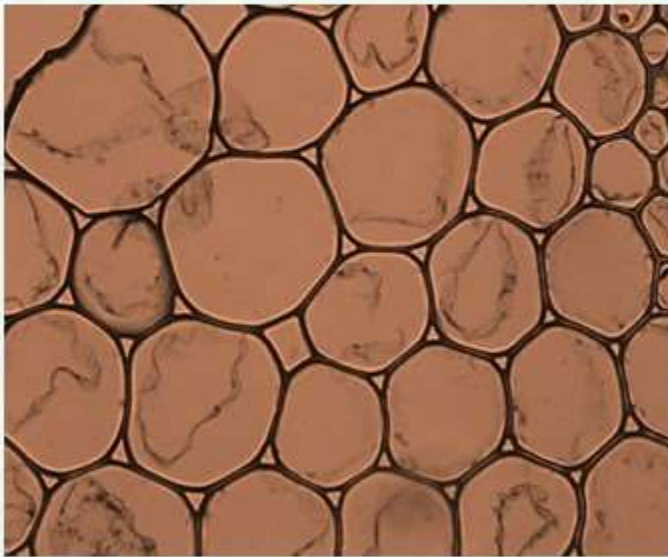
## SCLERENCHYMA

- Sclerenchyma cells are dead cells and they are devoid of protoplasm.
- The cells of sclerenchyma are closely packed without intercellular spaces.
- This tissue is present in stems, around vascular bundles, in the veins of leaves and in the hard covering of seeds and nuts. The sclerenchyma is mainly mechanical and protective in function.
- It gives strength, rigidity, flexibility and elasticity to the plant body and, thus, enables it to withstand various strains

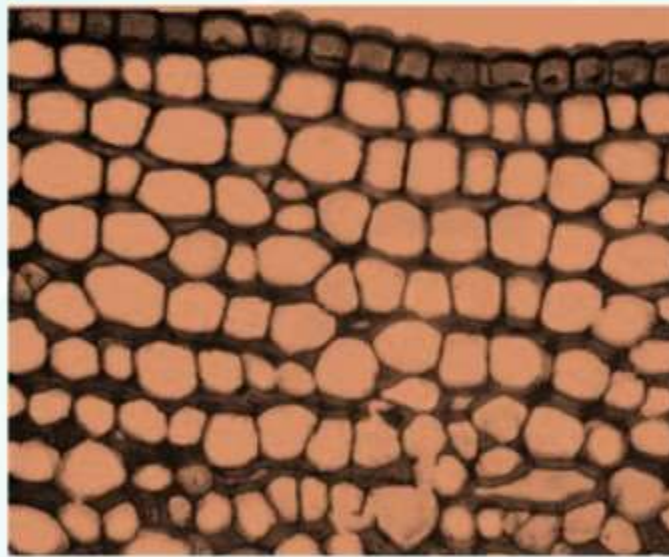


## SIMPLE PERMANENT TISSUES

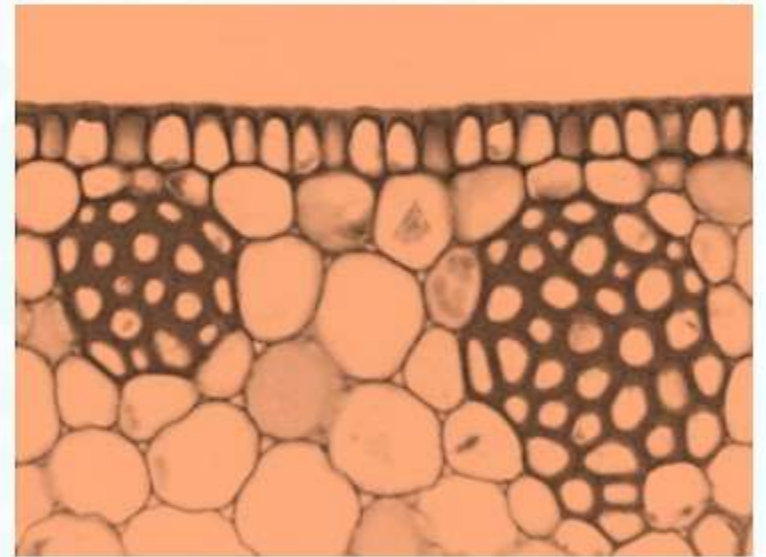
PARENCHYMA

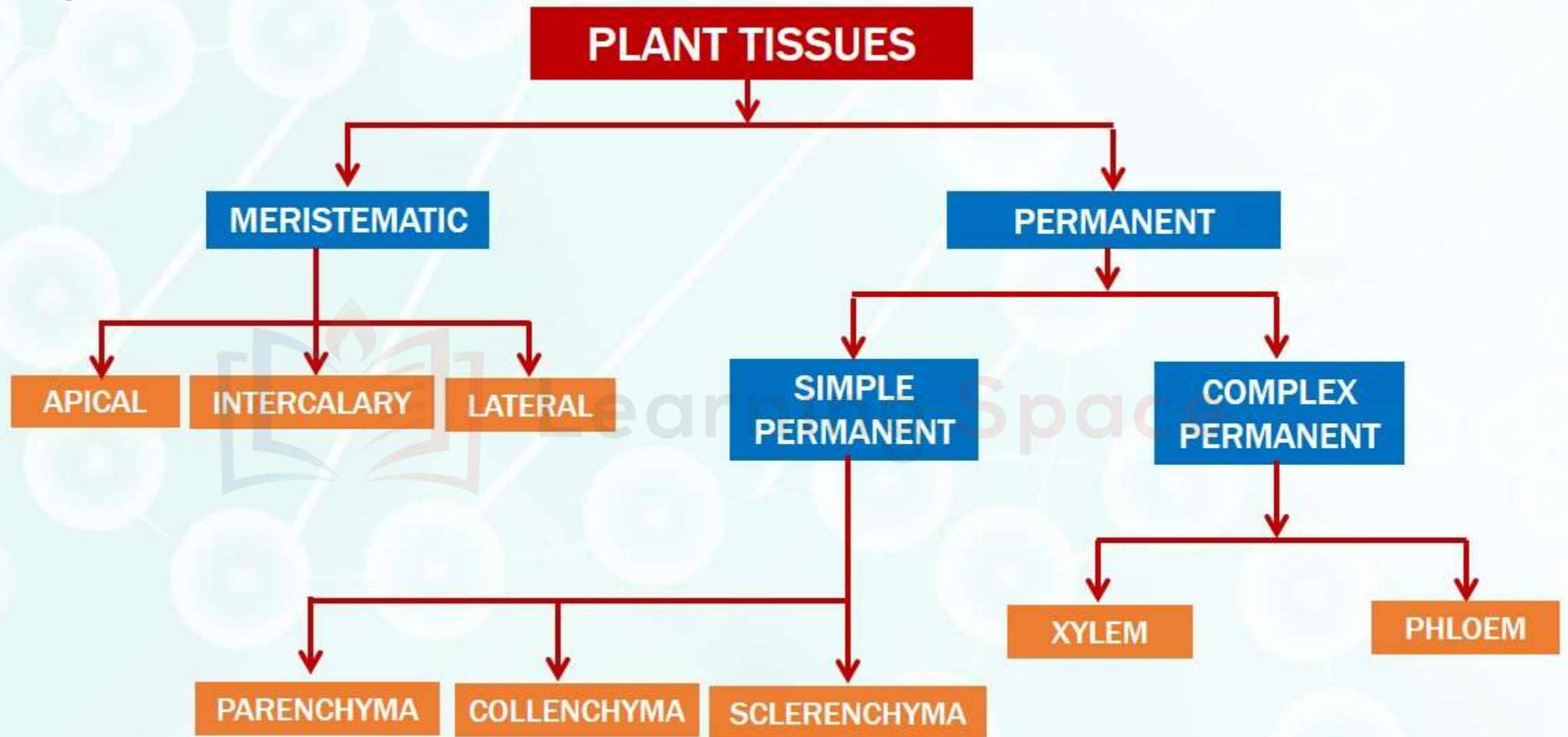


COLLENGHYMA



SCLERENCHYMA





## COMPLEX PERMANENT TISSUES



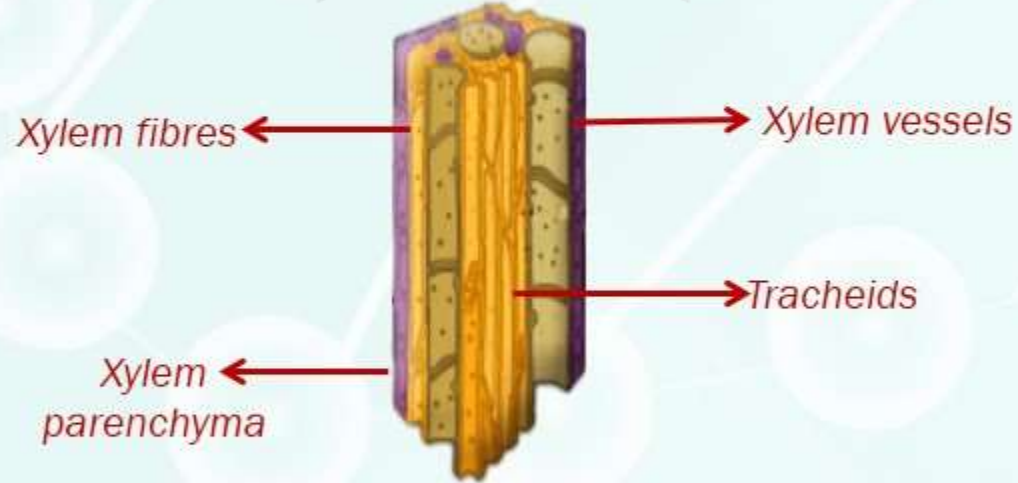
- The complex tissue consists of more than one type of cell having a common origin. All these cells coordinate to perform a common function.
- Complex tissues are of two types:  
**Xylem or wood and Phloem or bast.**
- Xylem and Phloem are both conducting tissues and also known as vascular tissues; together both them constitute vascular bundles.

*A vascular bundle is the transport system in vascular plants*

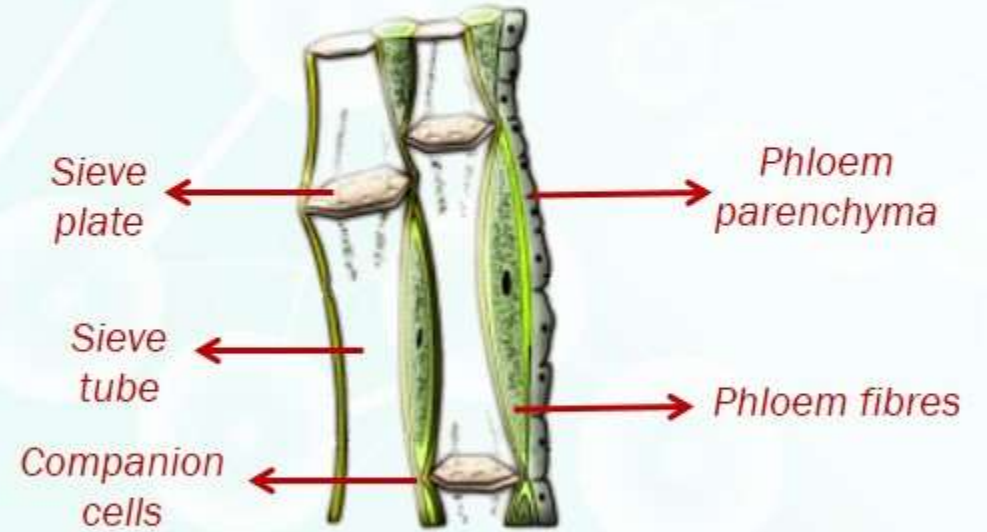


## COMPLEX PERMANENT TISSUES

### XYLEM



### PHLOEM



## COMPLEX PERMANENT TISSUES

### XYLEM

- It carries water and minerals from roots to the remaining parts of plant body.
- Functional xylem cells are non-living
- The movement in xylem is only in upward direction

### PHLOEM

- It translocates food from leaves to the remaining parts of the plant body
- Functional phloem cells are alive and living
- The movement of phloem is both in upward and downward direction



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